

SentinelGPT: An AI-Powered Large Language Model Framework for Advanced Cyber Threat Detection and Analysis

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1. Problem Statement

Modern enterprise cybersecurity operations face severe and escalating challenges in managing the overwhelming volume of network telemetry and security alerts generated by large-scale IT infrastructure. Security Information and Event Management (SIEM) systems process millions of event logs, network probes, and API requests every day across complex distributed cloud environments. Despite the scale of the threat landscape, many organizations still rely heavily on manual log inspection, static firewall rules, and fragmented monitoring tools — creating significant operational bottlenecks, severe analyst alert fatigue, and dangerously slow incident response times that allow malicious threats to propagate unchecked through internal networks.

One of the most critical challenges is identifying sophisticated multi-stage attacks such as credential stuffing, Distributed Denial of Service (DDoS) vectors, SQL injection payloads, and brute-force authentication spikes. Traditional rule-based intrusion detection systems generate extremely high rates of false positives while failing to correlate anomalous network behavior with established frameworks like MITRE ATT&CK. When a critical threat is detected, manual quarantine procedures require administrators to manually update firewall rules and isolate compromised IP addresses — a delayed process that exposes internal networks to lateral movement and unauthorized data exfiltration.

Existing security dashboards also lack transparent decision-making explanations, intelligent payload analysis, and real-time visualization capabilities. Security operators are frequently left without instant remediation guidance, making it difficult to understand why a specific threat score was assigned or what immediate countermeasures should be taken.

To address these critical challenges, this project presents SentinelGPT — an AI-Powered Large Language Model Framework for Advanced Cyber Threat Detection and Analysis. The system leverages intelligent autonomous software agents to automatically monitor network telemetry, calculate risk scores using heuristic anomaly algorithms, map threats to MITRE ATT&CK techniques, and autonomously quarantine high-risk IP addresses. By integrating real-time telemetry streaming, agentic quarantine execution, interactive analytics, a conversational AI triage assistant, and static payload scanning into a unified serverless cloud deployment, SentinelGPT eliminates response latency and delivers a scalable, reliable solution for modern enterprise cyber defense.

2. Difference: Traditional System vs. Proposed SentinelGPT

The following table highlights the key operational differences between the conventional manual security management approach and the proposed AI-powered SentinelGPT framework:

Traditional Security Operations System	Proposed Agentic AI — SentinelGPT System
Security alerts monitored manually using static firewall rules and basic log viewers.	Threats detected and analyzed automatically 24/7 via intelligent software agents.
High analyst alert fatigue; delayed incident detection and slow response times.	Real-time heuristic anomaly scoring (0-100) detects threats instantly with zero fatigue.
IP quarantine requires manual admin intervention and firewall rule configuration.	Autonomous Agent auto-isolates malicious IPs (score ≥ 75) in real time.
No standardized MITRE ATT&CK threat framework alignment or technique tagging.	All incidents automatically mapped to MITRE ATT&CK tactics and techniques.
Threat triage requires lengthy manual analyst research and verification.	Integrated AI Triage Assistant delivers instant remediation guidance.
Limited real-time visualization of threat velocity and perimeter metrics.	Live telemetry charts, risk gauges, and perimeter heatmaps on the dashboard.
Payload and log file analysis performed manually using external tools.	Built-in Payload Scanner automatically analyzes files with heuristic verdict reports.
Incident reporting requires manual log gathering and formatting effort.	Instant JSON incident-log export via REST API — one click.

3. Proposed Solution

SentinelGPT is an intelligent, full-stack, cloud-native cyber defense platform that automates threat detection, security monitoring, and incident mitigation. Unlike traditional reactive security tools, the system deploys autonomous software agents that continuously evaluate incoming network telemetry against heuristic scoring models and enforce security policies — replacing slow, error-prone manual workflows with instantaneous automated decisions.

The Telemetry Monitor Agent serves as the primary threat intelligence engine. It continuously analyzes packet rates, IP origins, payload signatures, and request patterns to compute a normalized risk score ranging from 0 (benign) to 100 (critical). Each detected incident is dynamically mapped to a corresponding MITRE ATT&CK technique such as T1078 (Valid Accounts), T1498 (Network DoS), or T1190 (Exploit Public-Facing Application) — and enriched with AI-generated remediation guidance specific to the attack pattern detected.

The Autonomous Quarantine Agent operates independently to enforce firewall isolation policies. When an incident risk score reaches or exceeds the critical threshold of 75, the agent immediately writes the offending IP address to the Blocked IPs database table, quarantining it from all further system access. Administrators retain complete manual override capability including revocation of quarantine entries and triggering simulated threat injections for testing purposes.

By unifying intelligent threat detection, automated quarantine enforcement, real-time telemetry visualization, conversational AI triage, and cloud-native serverless deployment into a single cohesive platform, SentinelGPT establishes a modern, scalable, and highly reliable solution for next-generation enterprise cybersecurity operations.

4. Overall System Architecture

SentinelGPT follows a modular, decoupled full-stack architecture. The security operator authenticates via the React 19 SPA frontend, which communicates with the FastAPI backend controller for telemetry snapshots and WebSocket streaming. Two autonomous agents process all incoming security events. All incident logs, quarantine records, and user credentials are persisted via SQLAlchemy ORM to SQLite locally or Vercel cloud storage in production:

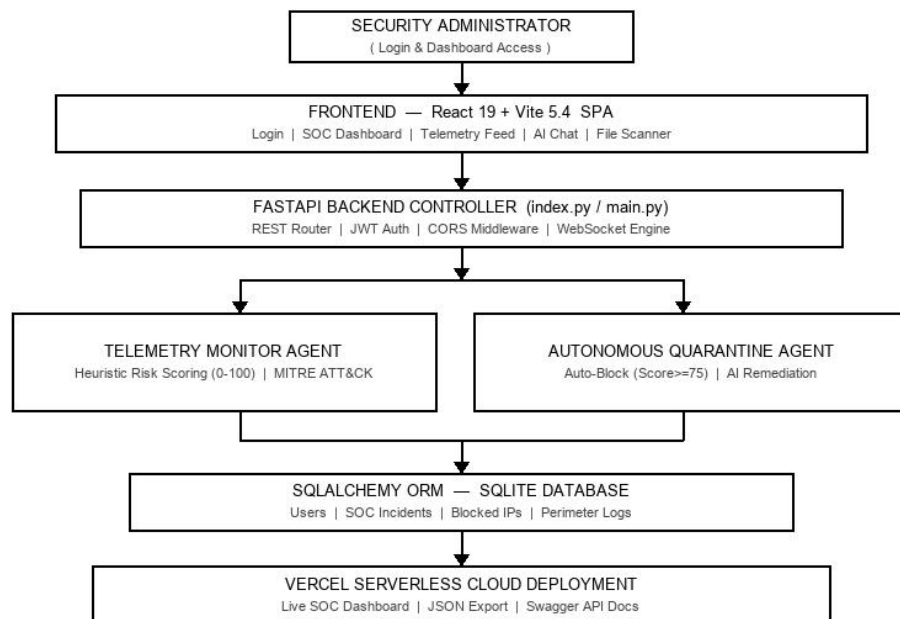


Figure 1: SentinelGPT System Architecture Diagram

5. Technologies Used

5.1 Frontend Technologies

React 19: Modern UI library for building dynamic glassmorphic dashboard components and managing real-time state updates.

Vite 5.4: Next-generation build tool providing ultra-fast HMR and optimized production bundles for serverless deployment.

Recharts: Data visualization library for rendering live threat velocity area charts, risk gauges, and severity donut charts.

Web Audio API: Browser-native alarm engine that fires audio alerts on Critical/High threat detection events.

5.2 Backend Technologies

Python 3.11+: Core language for implementing agent logic, threat heuristics, database ORM models, and server-side processing.

FastAPI: High-performance async web framework for RESTful API routing, WebSocket streaming, JWT authentication, and Swagger documentation.

Uvicorn: ASGI server powering local development execution and Vercel serverless function routing.

PyJWT: HS256 JSON Web Token generation and validation providing 8-hour expiring secure operator sessions.

5.3 Database & Deployment

SQLAlchemy ORM: Python Object-Relational Mapper providing type-safe, secure database interactions without writing raw SQL.

SQLite: Zero-configuration relational database for local development incident logging, quarantine records, and user accounts.

Vercel Serverless: Cloud platform serving the React static build and executing Python API functions at the network edge.

GitHub Actions CI/CD: Automated build and deployment pipeline triggered on every push to the main branch.

6. Architecture Flow & Agents Used

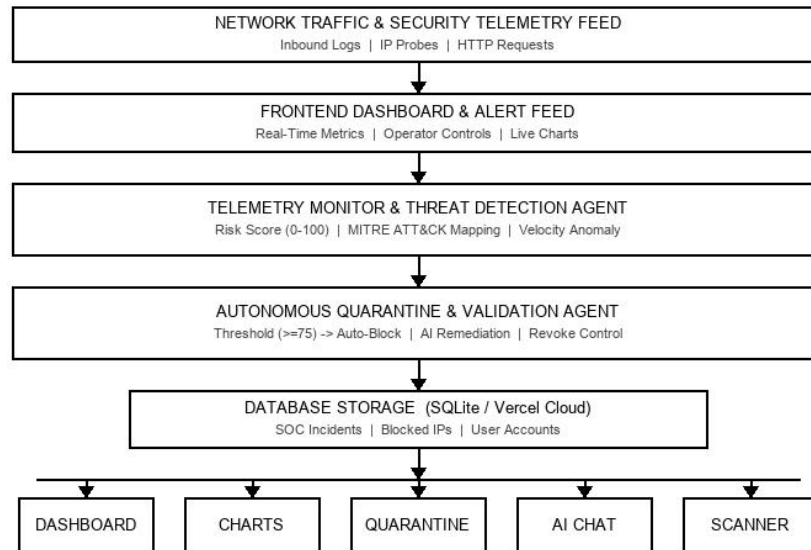


Figure 2: Agent Workflow — From Telemetry Ingestion to SOC Output

Agent 1 — Telemetry Monitor & Threat Detection Agent

Continuously inspects incoming network log events, extracts source IP details and payload metadata, computes a heuristic risk score (0–100), and assigns a MITRE ATT&CK tactic and technique tag. Produces a structured SOC Incident payload that is persisted to the database and streamed live to the frontend dashboard in real time via WebSocket.

Agent 2 — Autonomous Quarantine & Validation Agent

Evaluates every incident against the critical severity threshold (score ≥ 75). Qualifying incidents are automatically written to the Blocked IPs table, quarantining the offending source IP. The agent also validates manual revoke requests and generates contextual AI-driven remediation advice to prevent false-positive lockouts.

7. System Interface Screenshots

The following figures show the live operational screens of the SentinelGPT platform as seen by an authenticated security administrator:



Figure 3: SentinelGPT Real-Time SOC Operations Dashboard

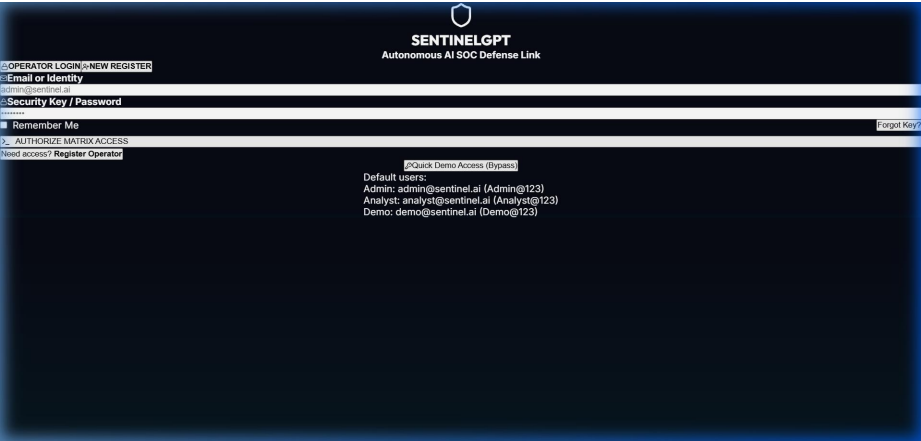


Figure 4: Operator Authentication Portal with Quick Demo Access

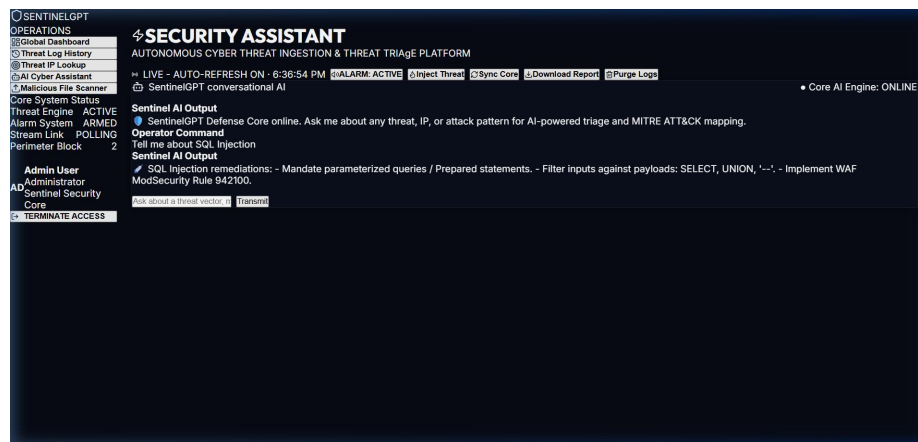


Figure 5: Conversational AI Threat Triage Assistant

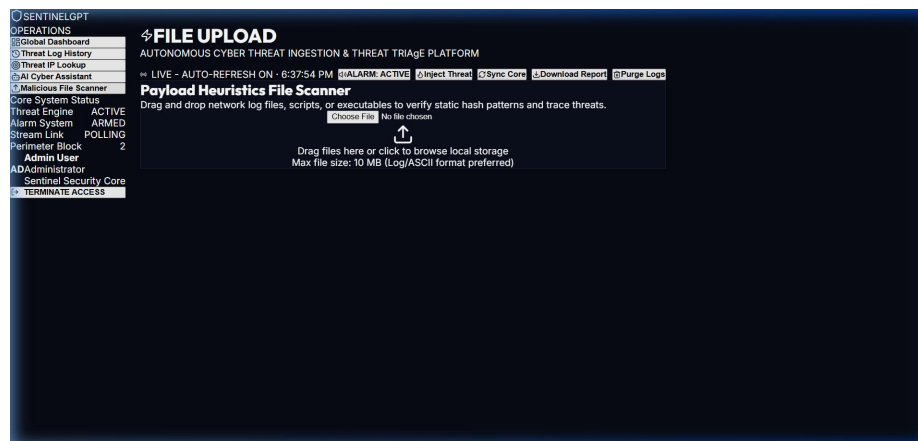


Figure 6: Heuristic Payload & Log File Scanner

8. Algorithms Used

1. Heuristic Threat Scoring Algorithm: Evaluates request velocity, payload anomaly patterns, and IP reputation to produce a normalized 0–100 risk score without relying solely on static signature databases.

2. Severity Matrix Classifier: Maps computed scores to four priority tiers — Critical (≥ 75), High (60–74), Medium (40–59), Low (< 40) — directly driving automated quarantine and alert escalation logic.

3. Velocity Rate Anomaly Detector: Monitors burst traffic spikes per source IP over sliding time windows to identify active DDoS attempts and port-scan sweeps in real time.

9. Implementation Phases

Phase 1 — Requirement Analysis: Identified SOC pain points, defined threat scoring metrics, mapped MITRE ATT&CK coverage requirements, and finalized system specifications.

Phase 2 — System Design: Architected the decoupled React SPA + FastAPI serverless backend; finalized REST API contracts, database schema, and agent workflow blueprints.

Phase 3 — Database Schema Design: Designed SQLAlchemy ORM models for Users, SOC Incidents, Perimeter Logs, and Blocked IPs tables.

Phase 4 — Frontend Development: Built glassmorphic React components including ThreatChart, RiskGaugeChart, BlockedIPs manager, AI Chat panel, File Scanner, and the Login portal.

Phase 5 — Backend API & JWT Auth: Implemented all REST routes (/api/snapshot, /api/block_ip, /api/unblock_ip, /api/sim_threat, /api/export, /api/login) with HS256 JWT authentication.

Phase 6 — Agentic AI Development: Programmed the Telemetry Monitor Agent (heuristic scoring + MITRE mapping) and the Autonomous Quarantine Agent (threshold enforcement + AI remediation).

Phase 7 — Integration & Testing: Injected simulated threat payloads to verify scoring accuracy, quarantine trigger reliability, manual override controls, and WebSocket streaming performance.

Phase 8 — Cloud Deployment: Deployed the complete full-stack application to Vercel Serverless; configured environment variables and verified live Swagger API documentation.

10. Applications

Enterprise SOC Operations: Continuous 24/7 network perimeter monitoring, automated threat triage, and instant IP quarantine for corporate security teams.

Financial Institutions: Protects banking portals and payment gateways from credential stuffing, brute-force attacks, and transaction fraud.

Cloud Service Providers: Safeguards serverless APIs, microservices, and distributed cloud infrastructure from unauthorized probes and exploits.

Healthcare Networks: Secures hospital systems and patient data endpoints against ransomware, malware, and PHI data exfiltration.

E-Commerce Platforms: Prevents DDoS disruptions and SQL injection attacks during high-traffic sales events, ensuring continuous uptime.

Academic & Research Institutions: Monitors campus networks, protects research data, and prevents faculty and student account hijacking.

11. Conclusion

SentinelGPT demonstrates that integrating Agentic AI with a modern full-stack serverless architecture — React 19, FastAPI, SQLAlchemy, and Vercel — can fundamentally transform reactive, manual security operations into a proactive, fully automated cyber defense system. By replacing human-dependent log review with autonomous, continuously running software agents, the platform eliminates analyst alert fatigue, reduces incident response latency to near zero, and provides uninterrupted 24/7 network perimeter protection.

The heuristic threat scoring engine, MITRE ATT&CK technique mapping, conversational AI triage assistant, and automated quarantine enforcement together establish a comprehensive, transparent, and scalable security operations platform suitable for deployment across enterprise, financial, healthcare, and academic environments. The project sets a strong technical foundation for the next generation of AI-powered cybersecurity frameworks.

12. References

- [1] FastAPI Documentation — <https://fastapi.tiangolo.com/>
- [2] React 19 Documentation — <https://react.dev/>
- [3] MITRE ATT&CK Framework — <https://attack.mitre.org/>
- [4] SQLAlchemy ORM — <https://docs.sqlalchemy.org/>
- [5] Vite Build Tool — <https://vitejs.dev/>
- [6] Vercel Platform Docs — <https://vercel.com/docs>
- [7] Python 3.11 — <https://docs.python.org/3/>
- [8] Uvicorn ASGI Server — <https://www.uvicorn.org/>
- [9] MDN Web Docs (HTML5, CSS3, JS) — <https://developer.mozilla.org/>
- [10] Russell, S., & Norvig, P. (2021). Artificial Intelligence: A Modern Approach (4th ed.). Pearson.
- [11] Wooldridge, M. (2009). An Introduction to MultiAgent Systems (2nd ed.). John Wiley & Sons.
- [12] OWASP Top Ten Security Risks — <https://owasp.org/Top10/>

13. Live Deployment & GitHub Repository

The SentinelGPT system is fully deployed and publicly accessible. Evaluators and reviewers may access the live working dashboard, interactive API documentation, and the complete project source code using the official links below:

Live SOC Dashboard	https://sentinelgpt-ai.vercel.app
Interactive Swagger API Docs	https://sentinelgpt-ai.vercel.app/docs
GitHub Source Repository	https://github.com/Pravallika2025/sentigraud-ai-.git